

## PART 4- SIMPLE LENGTH FILTERING EXPERIMENT

- **Tool used:** Filter sequences by length
- **Minimum length:** 10,000 bp
- **Input file:** Original genome assembly
- **Output:** Sequences  $\geq 10$  kb only

The screenshot displays the Galaxy web interface. The top navigation bar shows 'Using 5% of 268.4 GB' and the user 'suan\_jade'. The left sidebar contains navigation links: Upload, Tools, GalaxyAI, Workflows, Workflow Invocations, Interactive Tools, Notifications, and More.

The main panel is titled 'Filter sequences by length (Galaxy Version 1.2)'. It shows the 'Tool Parameters' section with the following settings:

- Fasta file:** 1: Hipposideros\_armiger\_genome\_original.fna.gz (as fasta)
- accepted formats:** (dropdown menu)
- Minimal length:** 10000
- Maximum length:** 0
- Setting to '0' will return all sequences longer than the 'Minimal length'**
- Additional Options:**
  - Email notification:** No
  - Attempt to re-use jobs with identical parameters?** No

The 'Output Tab' is visible on the right, showing a list of datasets. The top dataset is '6: Hipposideros\_armiger\_filtered\_10kb', which is highlighted. Below the tool parameters, the 'Additional Options' section is expanded, showing the 'Email notification' and 'Attempt to re-use jobs with identical parameters?' options.

The bottom panel shows the '6: Hipposideros\_armiger\_filtered\_10kb' dataset. It includes a 'Preview' button and a warning message: 'This dataset is large and only the first 100 KB is shown below.' The dataset is downloaded as a FASTA file. The preview shows the following sequence:

```
>NW_017731350.1 Hipposideros armiger isolate ML-2016 unplaced genomic scaffold, ASM189008v1 scaffold_0, whole genome shotgun sequence
AACGACAAGCTTGTGAATTATGGATTATGTAACGTGATTAATAAGTAATGccctttctccctccctctgCTCCCTG
GCGATTITTTGTGTTCTTGTCTCCCTCTGATCCAGTCCCCTCAATATTTACAACTGAGAGACACAATAACCTT
TATTTCTGTCAGATCAGAAGAGTCTCAGTGCAGAGATTAGGGTGATCGGGAATGCTCACCAGAAGATCCGAGGACCCCTG
GCAATCTCTCAGAGATCAGAGGGATGCAAGTTTCTCTAGAGGAAAGcaaccattatattattaggAGGCAATGTTTAA
AGAGCTAAAGTCTCAGTTCCCTGGACAGGACCGTTGTTGATTGTCAGCTTCTCTTAAAGAAAAGCACTGATTCTCAACAA
GAGACCGAGCTGCTGGCAGATAATGTCATGATGAGGTTGCGCTTTTACAGCTCGTGGATGGACTTGCAGCTGGGCTC
CTTTCCCGGCGCCACCAAGCTTCGATTGTTGGTTTGAAGTGAAGTCAAAACAGCAGGTGCAGCACTCCGAAGTCGGAATT
TCTTTAGTTGaatctcttctctccattCAAAGAGAAAATTAACCCAGCAATTAAGTCACTCTTTAGCTCTTC
TCAAGCAATACCTGGCACGGAGTATCAGACTTGACGTTAGATAAGGTAATGATGGGGAGGGGAAGTTGGAGAAATGAGAC
ACATGGTGCTAACATTTATCGCACATGattatgacaggtacttcACGgattatctcttattttaacctttacaAC
TGCAAAAGATGAGTATTTAATCTCTGTTTTCGAAATGATGCTGTGCGGGGTTACGTGACTTGTCAGATCGTACCCCTA
GTAAGTGGCTGAAGTGGGATTTGGTTCCCTTACAGGGCTTAGGTTGAACACCTCGAAGACCCCTCTCTGTTCCAAAGCT
TGTGCTTCTTCCACGCTAtataagggaataatttttaaaaaagacacttggtggccgcccgaagtggtcagtt
ggtaagagctcaatcaagggtcgcgctgagcaacagggttcacccggttcggtccgccaatccgagctgccaac
aggtgacagatgctcagctaaatagaaaaaaagaaagaaagacattggTTATAATTTCCAGAGGCTCA
CGGAGCAGGATGAGAATCCAGTTGTGCTCGTATCCATCTGCTCTCAGCTATGTCACCAATTCCTTGTCTGCTCTTC
TTCCTCGGTTCTCCTGTTTACCCCTGCGGTCATAGTGAGATGGATGCTTGAactttatctctctctcttctgaggTG
AGAGTGGTGAGGATGAGGACAGTGGTGAAGCAGTACAGCATCAAGTCTTAATTTGACGGGCACTCTGTGAGCTTTTC
```

**Figure 4a.** Length-filtering experiment showing tool parameters (minimum length = 10,000 bp) and the resulting filtered genome output. Sequences shorter than 10 kb were removed to evaluate their contribution to total assembly size.